Problem 2 b)

What is the pdf of the radius of 2 $\mu = 0$, $\sigma = 1$ Gaussians? X_1, X_2

Gaussian distribution general form

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}} \qquad f_{X_1}(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} = f_{X_2}(x)$$

Note $\mu = 0$ $\sigma = 1$ $\equiv$ "Standard Normal Distribution"

$$\frac{f_{X_1}(x)}{f_{X_2}(x)} = \frac{\frac{1}{\sqrt{2\pi}} \cdot e^{\frac{-x^2}{2}}}{\frac{1}{\sqrt{2\pi}} \cdot e^{\frac{-x^2}{2}}} = \frac{\sqrt{2\pi}}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} \cdot e^{\frac{x^2}{2}} = 1 \cdot e^0 = 1$$

Flat distribution
Constant Probability

Instead

If $f_{X_1}(x) \neq f_{X_2}(y)$

$$\frac{f_{X_1}(x)}{f_{X_2}(y)} = \frac{\sqrt{2\pi}}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} \cdot e^{\frac{y^2}{2}} = e^{\frac{y^2 - x^2}{2}}$$ which?

I'm assuming the answer being searched for here is 1 ~ the constant probability, 2 "identical" normal distributions

c) 2D Gaussian $a + bi \Rightarrow \sqrt{a^2 + b^2} \equiv (|a^2| + |b^2|)^{1/2}$? $=$

The

$$a = \frac{1}{\sqrt{2\pi\sigma_a^2}} e^{\frac{-(x-\mu_a)^2}{2\sigma_a^2}} \qquad b = \frac{i}{\sqrt{2\pi\sigma_b^2}} e^{\frac{-(x-\mu_b)^2}{2\sigma_b^2}}$$

$$|a^2| = \frac{1}{2\pi\sigma_a^2} \left(e^{\frac{-(x-\mu_a)^2}{2\sigma_a^2}} \right)^2 = \frac{1}{2\pi\sigma_a^2} e^{\frac{-(x-\mu_a)^2}{\sigma^2}}$$

$$|b^2| = \frac{1}{2\pi\sigma_b^2} \left(e^{\frac{-(x-\mu_b)^2}{2\sigma_b^2}} \right)^2 = \frac{1}{2\pi\sigma_b^2} e^{\frac{-(x-\mu_b)^2}{\sigma^2}}$$

$$\frac{1}{2\pi\sigma_a\sigma_b}$$

434 Homework 3 Stelter

48 Birds
52 People

1st Algorithm

	Real Birds	Real People
Birds	45	5
People	3	47

Total Error: 8 (8%)
Preferred Error: 3
(Real Birds => People)

2nd Algorithm

	Real Birds	Real People
Birds	47	11
People	1	42

Total Error: 12 (12%)
Preferred Error: 1
(Real Birds => People)

1 a The first algorithm made fewer mistakes (8 < 12)

b The second algorithm is better for the zoo by virtue of the fact that it allowed for only 1 bird to escape.

c 99 Truth - 99 Birds, 1 Person or 98 Birds, 1 Person

1st Algorithm

Birds / People	Actual Birds	Actual People
	90	

Classification Behaviors	A	B
Bird ✓ Bird	45/48	47/48
Bird x Person	3/48	1/48
Person x Bird	5/52	11/52
Person ✓ Person	47/52	42/52

e continued

New Population of 100 (1 Person)

A (Predicted)

	Real Birds	Real People
Detected Bird	92.81	.0961
Detected Person	6.1875	.9038

B (Predicted)

	Real Birds	Real People
Detect Bird	96.94	.212
Detect Person	2.0625	.808

Problem 2

a) Sum of 2 Identical Exponential Distributions

PDF of Two Distributions Summed = convolution of their pdfs

$PDF(2 \cdot \lambda e^{-\lambda x}) \Rightarrow \int_{-\infty}^{\infty} f_1(x) f_2(z-x)\, dx = PDF(z)$

$f_1 = \lambda e^{-\lambda x}$ for $x \geq 0$

$f_2 = \lambda e^{-\lambda x}$ for $x \geq 0$

$\Rightarrow f(z) = \int_0^z f_1(x) f_2(z-x)\, dx \quad z \geq 0$

$\Rightarrow \int_0^z \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(z-x)}\, dx = \lambda^2 \int_0^z e^{-\lambda x - \lambda(z-x)}\, dx$

$= \lambda^2 \int_0^z e^{-\lambda z - \lambda x + \lambda x}\, dx = \lambda^2 \int_0^z e^{-\lambda z}\, dx \Rightarrow \begin{cases} \lambda^2 e^{-\lambda z} \cdot z & z \geq 0 \\ 0 & z < 0 \end{cases}$

This is a gamma distribution